



Competencies for the artificial intelligence age: visualisation of the state of the art and future perspectives

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Abstract

The objective of this paper is to perform an in-depth analysis of the literature on the competencies for implementing and leveraging artificial intelligence (AI) within organisations. From a bibliometric study using SciMat with articles from the Web of Science database, we identified 421 papers published between 1992 and 2020. This study offers a systematisation of the competencies and skills for AI, highlighting the most prominent, basic, specialised and emerging themes, and providing a performance measure analysis of this field. In addition, major challenges and a research agenda are discussed. The organisational challenge is to achieve a workforce with the necessary digital competencies, and to adapt human resource management practices to AI challenges.

Keywords SciMat · Artificial intelligence · Digital competencies · HRM · Future of work

JEL Classification M50

1 Introduction

Today, due to the rapid development of artificial intelligence (AI) and robotics, we are on the cusp of the next industrial revolution, known as “Industry 4.0”. Revolutionary changes are emerging (Strandhagen et al. 2017) through the use of smart technologies that enable new and more efficient processes, products and services (Lee et al. 2018; Zhong et al. 2017). This change presents risks, challenges and

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opportunities for industry stakeholders and for communities (Hirsch-Kreinsen 2016; Roblek et al. 2016; Shamim et al. 2016; Strandhagen et al. 2017).

Such changes will affect both companies and society, although in different ways (Rajnai and Kocsis 2017), since society will react more slowly, remaining in a constant stable situation. In the case of companies, one of the main difficulties they face is to integrate AI and HRM management issues, due to the lack of employee competencies and skills to promote radical changes throughout the organisation (Gowen and Tallon 2003). Outplacement activities, developing new business models, reinvestments in new industrial products and equipment and new services also change the employment scenario (Berger and Frey 2016). Some jobs will disappear and be replaced, while other, completely new jobs will emerge. The most demanded professions at present will not exist in the next five or ten years, and 65% of children who are currently in primary school will work in completely new jobs that do not currently exist (Rampersad 2020).

Labour limitations and the number of jobs are only the tip of the iceberg; in a broader perspective, the role of the human factor will be indispensable in smart industry. This consideration does not only influence production, but also the occupational dimension, employment dynamics, working conditions, education systems and the demand for competencies and national policies (Benešová and Tupa 2017; Rajnai and Kocsis 2017). Specifically, qualifications and competencies will be critical as companies are obliged to adapt to constant technological changes (Harkins 2008), making knowledge management (Nonaka et al. 1996) and knowledge of information technologies (IT) (Pfeiffer 2016) the keys to success for highly innovative companies (ITU 2016; Gehrke et al. 2016). Indeed, according to Kindermann et al. (2020), certain competencies are the key to understanding why some organisations can create value from ubiquitously available digital technologies; digital competencies are crucial to a company's digital strategy.

Therefore, the first challenge is to analyse the links between AI and employees' competencies; competencies will become a key factor, especially in knowledge-intensive jobs in the field of computing, self-study, algorithms and data analysis (Benešová and Tupa 2017). This new reality demands that companies seek to fill the gaps in their dynamic capacities (Eisenhardt and Martin 2000) to adapt and survive in the new competitive reality, combining real and virtual global information and IT management knowledge (Hecklau et al. 2016). Indeed, organisations are trying to determine what role their resource types play in value creation in this new digital context (Boukis and Kabadayi 2020). All of these terms are used to describe the management and market revolution that results from Industry 4.0. Appropriate qualifications, competencies and learning systems are relevant issues to support the future agenda of organisations. In a scenario in which the relationship between AI and employee competencies is being debated, high levels of excellence can be achieved.

In effect, according to Gallardo-Gallardo and Collings (2021), technological changes have reduced competency cycles; skills that were crucial a decade ago are no longer valuable, while many new competencies will emerge to meet the needs of tomorrow's jobs, and talent needs will have to be forecast. Additionally, in line with Gallardo-Gallardo and Collings (2021), Shamim et al. (2016) identify a lack of

skilled labour for Industry 4.0. Thus, the study of what digital competencies entail is highly relevant, and is the purpose of this research. The main research questions of this study are: how has the relationship between employees' competencies and AI evolved? What employee competencies are required to implement and leverage artificial intelligence within organisations? What implications will this relationship between competencies and AI pose to HRM? Indeed, the aim of this paper is to address the relationship between AI and digital competencies by focusing on the HRM side. The main contributions of this work lie in the original results resulting from a bibliometric analysis through the identification, classification and analysis of current knowledge on the relationship between employees' competencies and AI. We developed an original research agenda to expand what is still a scant discussion on the competencies for implementing and leveraging AI within organisations.

Following this introduction, section two reviews previous research about relationships between AI, digital competencies from employees and HRM. Section three explains the data collection process and the bibliometric analysis conducted using the SciMAT science mapping tool. Section four shows the performance bibliometric study carried out, including the selected papers considered in this research and other bibliometric data, such as citations, and the impact of authors, journals and countries. Section five presents the findings of the science mapping study. Finally, the paper ends with the main conclusions, the managerial implications, limitations and a research agenda.

2 Theoretical background

2.1 AI, digital competencies and HRM

AI was established as an academic discipline in the 1950s, yet it remained an area of limited academic and practical interest during the second half of the twentieth century. Currently, due to the improvements in computing power, AI has entered into business environments and conversations of all kinds with many associated terms such as machine learning or big data. Artificial intelligence is a fuzzy concept, mainly due to the different definitions of intelligence itself and to the variety of AI definitions (Kaplan and Haenlein 2019). According to Kaplan and Haenlein (2019), artificial intelligence is the system's ability to understand and learn from external data, and to utilise it to perform particular tasks and meet objectives through versatile adaptation, while machine learning is an important but limited part of AI. AI uses emerging technologies that enable machines to perform tasks in a similar way to humans in areas such as cognition, sensing and performing (Akerkar 2019; Malik et al. 2020a, b). In turn, big data refers to technologies that allow the collection, management and analysis of data sets that are too large for conventional database systems.

An important challenge for organisations is to align AI related issues with their employees, including HRM practices (Malik et al. 2020a, b), together with areas such as competency development (Kumar et al. 2018). The reason is that the rapid development and use of AI in organisations implies the substantial modification of

an important variable from the human resource perspective: workers' competencies. In relation to competencies, there is evidence that digital transformation alters the characteristics of work and modifies the demand for knowledge, skills and behaviours, giving priority to people with both digital and non-cognitive competencies (Gonzalez-Vazquez et al. 2019). It is therefore necessary to consider whether AI alters or modifies the competencies that employees require within organisations which embrace AI.

Although digital technologies are the basis for innovation, they do not generate it by themselves. Innovation is carried out by people, so human capital becomes a key competitive factor (Díaz-Fernández et al. 2014; van Laar et al. 2017), according to the resource-based view (RBV) (Barney 1991). Under the RBV, human capital is identified as a key factor in the attainment of a sustainable competitive advantage (Barney 1991), since competencies explain the differences between competitive advantages in human resources (Hayton and McEvoy 2006).

A competency describes a combination of skills, attitudes and behaviours that an individual possesses or that the organisation expects of a particular employee (Hayton and McEvoy 2006). Although there are multiple types of competencies, in this situation, digital competencies become a key human capital resource (van Laar et al. 2017). Digital competency is defined by the European Commission in the DIGCOMP project as *the set of knowledge, skills, attitudes (thus including abilities, strategies, values and awareness) that are required when using ICT and digital media to perform tasks; solve problems; communicate; manage information; collaborate; create and share content; and build knowledge effectively, efficiently, appropriately, critically, creatively, autonomously, flexibly, ethically, reflectively for work, leisure, participation, learning, socialising, consuming, and empowerment* (Ferrari 2012, p.3). In addition, digital competencies can be defined as *a set of basic knowledge, skills, abilities and other characteristics that enable people to efficiently and successfully fulfill their job tasks with respect to digital media* (Oberländer et al. 2020, p.5). Digital competences arise as a consequence of technological changes that create new tasks and new roles both in society and in business management. Digital skills are those that contribute to the cultural and digital transformation of companies due to the incorporation of new technologies into business strategies, projects and processes. They are not only technological skills but also the acquisition of knowledge, values, attitudes, ethics and regulations in digitisation processes.

It is relevant to highlight how AI changes the role of HR in companies (Malik et al. 2020a, b). HRM is characterised by a high level of complexity (e.g., measuring employee performance), which has major consequences for both employees and the company (Tambe et al. 2019). Moreover, different types of tasks are developed by employees, essentially, mechanical tasks (e.g., repair and maintenance of equipment), thinking tasks (e.g., processing, analysing and interpreting information) and feeling tasks (e.g., communication with people) in different categories of jobs. Huang et al. (2019) empirical study shows that, in the future, human employees will increase their presence in positions with feeling tasks, while thinking tasks will be performed by AI systems, similarly to how mechanical tasks were taken over by machines and robots.

In this context, the HRM practices traditionally followed in organisations need to be modified. Changes have been observed in recent years in HRM, especially related to planning, recruitment and selection, remuneration, performance appraisal, employee relations, culture, health and safety, training and development, studying the advantages and disadvantages of digital transformations, developing new employee competencies and assessing performance (Fenech et al. 2019). Moreover, recruitment and selection practices must incorporate those candidates with the requisite competencies and the highest potential, training and career development must favour continuous learning for employees and help them to adapt to the demands imposed by technological developments.

In addition, the changes caused by digital technologies and innovations, such as big data, autonomous robots, IoT, clouds, virtual reality and virtual intelligence, necessarily imply training and adapting the workforce. However, some of the traditional competencies such as emotional intelligence, creativity, flexibility and managing others cannot be performed by machines. For this reason, the need to attract workers who are active, adaptable and able to quickly accept new ideas and responsibilities becomes especially relevant (Cantoni et al. 2018). The human resource management practices most influenced by technology at the moment are: recruitment, training and development, resource allocation, internal communications and talent management (Malik et al. 2020a, b). For example, recruiting (i.e. e-recruiting) through social networks is already an innovative and cost-efficient reality (Böhmová and Chudán 2018). People who use social media for recruitment need to know how a social network is used and what kind of data can be obtained, as well as understand the principles of the social network (Mazurchenko and Maršíková 2019). In this line, Malik et al. (2020a, b) indicate that AI applications used to perform some HRM tasks contribute to HR cost effectiveness and improved employee experience in multinational companies. However, these authors also warn of the ethical issues involved in this digital technology context.

In summary, the above arguments suggest that AI and other technological developments have modified the set of personal competencies demanded by organisations. In particular, digital competencies and non-cognitive competencies are now required in order to realise the full potential of AI while also having the competencies to perform the tasks that machines cannot do. In addition, AI has also changed HRM practices in two directions: on the one hand, they must support the development of the new competencies (digital and non-cognitive) demanded by the organisation while, at the same time, AI is contributing to the implementation of some of these HRM practices.

To conclude, it seems clear that the development of AI is going to condition the way companies operate, making it an interesting topic for an in-depth study on the competencies that will implement and leverage artificial intelligence within organisations and, therefore, on human resource management and how employees can contribute to organisational success through the skills they bring to the organisation. To this end, a bibliometric study was conducted to explore these relationships.

3 Method

3.1 Methodology

In order to know the current state of the literature on the competencies for AI and how this increasingly relevant topic emerged and evolved, we performed a bibliometric analysis using the SciMAT science mapping tool, developed by Cobo et al. (2012). Bibliometric methodology is the statistical study of scholarly documents (Garfield 1955) and enhances the comprehension of the intellectual architecture of a scientific field with objectivity (Garfield 1979). Bibliometric studies conducted by means of science mapping offer profound analyses, highlight the intellectual structure and detect relevant topics (Cobo et al. 2011b; Deng et al. 2020). SciMAT was selected as it provides most of the advantages of the current science mapping software techniques and because it is based on a robust and concrete method grounded on bibliographic networks and bibliometric indicators (Cobo et al. 2011a). According to Cobo et al. (2012), SciMAT allows researchers to visualise maps by using co-word analyses with a longitudinal perspective that provides information on the research themes in an academic field of study, as well as by tracking the evolution of the scientific work along different periods.

The following workflow was applied to conduct the science mapping analysis (Cobo et al. 2012; Paule-Vianez et al. 2020): (1) data collection, (2) data improvement, (3) network creation and normalisation, (4) map construction, (5) analysis and visualisation, and (6) performance analysis (see Fig. 1 for an overview of the

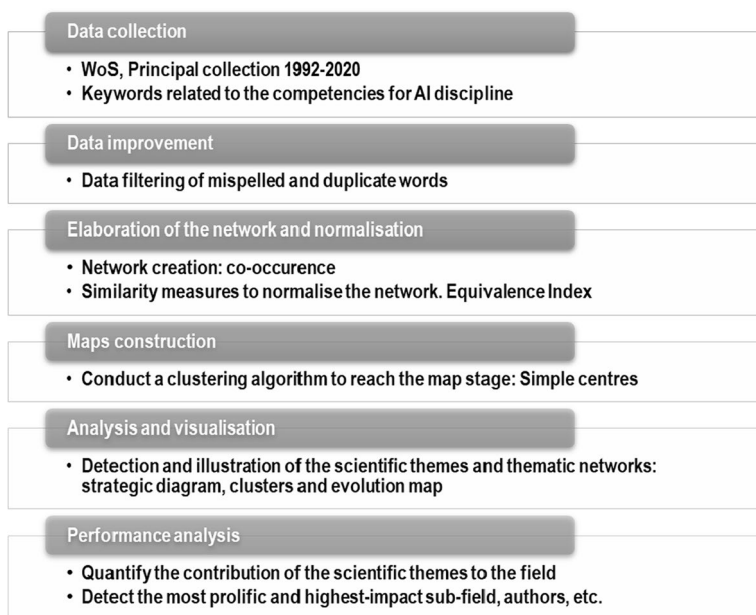


Fig. 1 Science mapping workflow

workflow). Steps 1 and 2 are explained in the following subsection (“Data”) of this methodology section, to provide separated and detailed information about the data. To create the network and normalisation, co-occurrence selection was performed to obtain the network, and the equivalence index was used to normalise the network. In the map construction, the simple centres algorithm allowed us to identify themes or clusters.

In the analysis and visualisation, the academic themes and thematic network identified were graphically represented using two instruments: a strategic diagram and a thematic network. We considered two dimensions to represent each theme in the strategic diagram: centrality and density. Centrality can be interpreted as the relevance of the research theme to the field of study, whereas density is considered as an indicator of the development of the topic. Considering these two dimensions, four quadrants were visualised in the strategic diagram (Fig. 2): motor themes (well-developed and important for the scientific field), basic or transversal themes (relevant to the field but not fully developed), specialised or isolated themes (well-developed but marginally relevant to the field), and emerging or declining themes (both poorly developed and marginally relevant) (Cobo et al. 2012; Paule-Vianez et al. 2020). In the strategic diagram, the scientific themes are represented by spheres, where the volume is proportional to the number of related documents.

In this step, we identified and analysed the evolution of the scientific topics over separate periods to uncover the evolution of the main thematic areas, their roots and connections. In the evolution map, the continuous lines represent linked topics with the same name, so either the topics are defined with the same keywords or the label

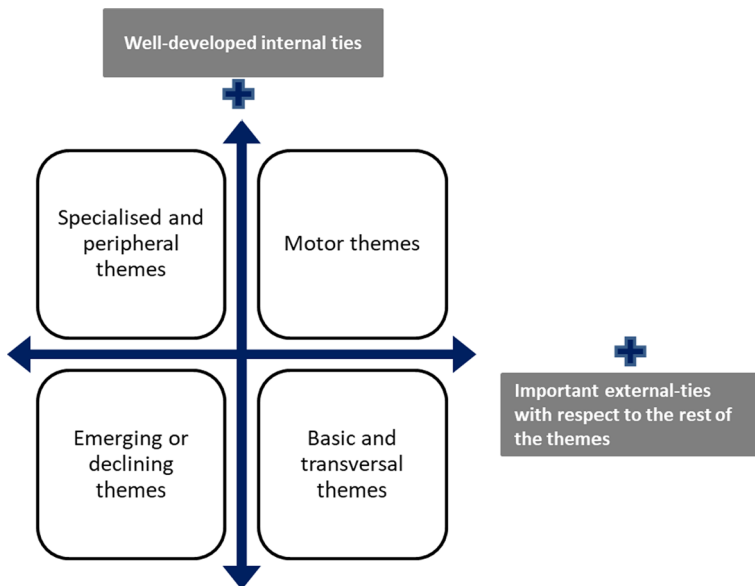


Fig. 2 Strategic Diagram

of one topic is part of another, whereas the dashed lines show that the topics share characteristics other than the topic name (Alcaide-Muñoz et al. 2017).

Lastly, in the performance analysis, we measured the performance of the research themes. To identify the most prolific scientific domains with the greatest impact, their contribution to the field of study was qualitatively and quantitatively measured using the following bibliometric indicators: the number of published documents, the number of citations and the *h*-index.

3.2 Data

The data collection process and data improvement are described in this subsection. As mentioned above, this study conducted a co-word analysis to analyse the conceptual structure and the main research topics related to AI and competencies. Co-word analyses retrieve relationships among concepts that co-occur in document titles, keywords or abstracts (Callon et al. 1991; Martínez et al. 2015; Zupic and Čater 2014). Co-citation analyses use co-authorship data to assess collaboration (Zupic and Čater 2014). Maps created using co-word analyses in a longitudinal framework provide information on the themes in a scientific field and can be used to analyse and track consecutive time periods in the evolution of an academic discipline (Garfield 1994). Therefore, the key data for this study are the concepts/keywords used in the document title, author and journal keywords, and abstract.

This manuscript analyses articles and reviews written in English in Management and Business categories from the Web of Science (WoS) Core Collection database, as this study focuses on the analysis of the employees' or leaders' competencies needed to leverage AI. WoS provides complete and detailed information about each document (author, title, source, abstract, keywords, institution, citations, etc.). The search was performed in June 2020 to obtain the most updated information about competencies for AI. After consulting two experts in the AI research field, and based on our expertise in the HR field and our awareness of relevant articles already published, two selection criteria were applied: first, that the articles had one section covering AI related keywords within organisations, including the keyword *twenty-first-century*, as it is a recurrent keyword amongst business researchers to refer to the future organisational challenges derived from technological developments; and second, that they had another section on competencies and HR related terms. The combination of these two sections searched for articles and reviews related to AI and competencies. In addition, to avoid missing any relevant articles, the search criteria scanned any document containing the most common keywords used by researchers in the field of AI and competencies, i.e., *digital competencies* and *digital skills*, combined with general and broad HR related keywords. By introducing this combination of keywords, we attempted to narrow down and focus the search on our target: competencies for the AI field from the HRM perspective. Selecting the criteria for keyword inclusion is an important part of a bibliometric study. This bibliometric study is focused on the research on competencies for AI, so the selection of the correct keywords is crucial, as it is important to avoid missing any relevant combination. Table 1 shows the keywords that were selected to conduct this science

Table 1 Competencies and skills for AI application keywords used to search in the WoS

TS=((("big data" or "artificial intelligence" or "machine learning" or "deep learning" or "data scientist" or "data science" or "visual analytics" or "people analytics" or "HR analytics" or "21st-century*" or "twenty first century") and ("competenc*" or "skill*" or "e-skill*" or "digital literacy" or "human resource management" or "HRM")) or (("digital competenc*" or "digital skill*") and ("human resource*" or "employ*" or "workplace*" or "workforce*")))

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mapping analysis. Using the WoS, this analysis searched for all articles and reviews containing these combinations of keywords in the title, abstract and keywords sections, generating 421 documents. Next, the retrieved information was uploaded to the SciMAT software, where the data were analysed. To improve the data quality, the study started with 1988 words, of which 110 words were reduced to 55, as they were in both plural and singular forms and therefore represent the same concept. Next, we examined the data to detect incorrect, repeated or misspelled words (i.e., terms representing the same meaning such as HRM or human resource management were merged as human resource management; spelling variations such as behaviour and behavior were merged, and so on). Additionally, words with a very general or no meaning were excluded (i.e., impact, framework, model, systems, etc.). In total, the study considered 1874 words.

Regarding the period, since competencies for AI is a novel field of study, the number of publications in the early years was small. The first two decades, in which fewer documents were published, were considered for the earliest period, that is, from the first year that the SciMAT tool detected relationships among the data, 1992, until 2010. The last decade (2011–2020), as the discipline developed, was divided into two periods: 2011–2015 and 2016–2020. The two first periods (1992–2010 and 2011–2015) yielded 72 and 87 published documents, respectively. Prior to 1992, the SciMAT tool revealed no strong interrelations between the thematic networks. Lastly, the most recent period considered, 2016 to mid-2020, yielded 262 published documents. With this separation, the evolution of competencies for AI can be analysed from its beginning to the present time.

4 Performance bibliometric study

This section offers a description of the AI and competencies field of study considering publications, citations and impact by examining the following bibliometric indicators: published articles, citations received, journal impact factor, data on the geographic distribution of publications, *h*-index and most cited papers and authors. First, the bibliometric performance study analysed the production and impact of published documents and, second, the production and impact of authors, journals and countries.

4.1 Publications and citations

Figure 5 (see “Appendix”) displays the number of publications by year related to competencies for AI, showing the growing trend. During the first 18 years, there was a slight upward trend: below 10 documents each year from 1992 to 2010, and below 20 documents each year from 2011 to 2015. Therefore, for this analysis, this initial stage was separated into two parts: 1992–2010 and 2011–2015. The 2016–2020 period was the most prolific for the AI and competencies field. In the years 2016 and 2017, the number of publications exceeded, for the first time, 20 and 60 documents, respectively. Accordingly, the number of published documents (articles and reviews) from 1992 to 2010, 2011 to 2015 and 2016 to 2020 was 72, 87 and 252, respectively.

The distribution for the citations is more erratic, although the trend is positive overall. This can be explained by the novelty and initial stage of this field of study, and the impact of AI on requisite competencies. A total of 4732 citations were recorded from 1992 to mid-2020. Moreover, according to the WoS, the average citation rate per article is 11.46; thus, given this pattern of progression, a positive tendency can be expected.

4.2 Production and impact of authors, journals and countries

To further our understanding of this field of study, we identified the most prolific authors, cited documents and journals. The most prolific author is Harvey, with three publications, followed by the rest (i.e., Akhtar, Baporikar, Bondarouk, Boudreau, Cascio, etc.), with two publications each in this field in the WoS.

Table 4 (see “Appendix”) presents documents with more than 50 citations in the WoS. The first column indicates the publication title, the second the authors’ names, the third the publication year and the fourth the number of citations of this publication. The most cited publication is by Waller and Fawcett (2013) about artificial intelligence and supply chain management, with 369 citations since 2013. One of the most recent publications with the highest number of citations is Huang and Rust’s (2018) paper on artificial intelligence in services, with 92 citations since 2018.

Table 5 (see “Appendix”) lists the most prolific journals in this field of study. The International Journal of Human Resource Management, Human Resource Management Review, Management Decision and Human Resource Management are among those with the highest number of publications (at least four documents) about competencies for AI. The first column indicates the journal name, the second the number of publications on this topic in the journal according to the WoS, and the third, the five-year impact factor. As can be seen, a broad range of journals has accepted publications in this field, which reflects the increasing interest in the topic from well-reputed journals.

Finally, Table 6 (see “Appendix”) shows the most prolific countries in terms of published documents, considering the authors’ affiliations. Documents with authors from institutions in various countries are assigned to several locations, so the total number of publications by country is larger than the total number of documents on AI and competencies, i.e., 550. Undoubtedly, the United States has the largest number of publications in this field, with 149 documents from 1992 to 2020, followed by the United Kingdom, with 66 (i.e., 27% and 12%, respectively, of the total number of publications). Other

countries such as Canada, Australia, India, the People's Republic of China and Germany are increasingly interested in researching this discipline, presenting a modest but stable growth trend across the periods analysed.

5 Science mapping study

5.1 Content analysis

As we have seen, the number of publications related to AI and competencies is increasing. In total, 421 articles and reviews have analysed the competencies and skills for AI. This section studies the research themes studied in each period, with a special focus on the most prolific period: 2016–2020.

5.1.1 1992–2010

In the period 1992–2010, *learning* is a motor theme (well-developed and important for the structure of the discipline) (Fig. 6, see “Appendix”). The *learning* research theme covers subthemes such as *knowledge* and *education*. Lepak and Snell (1998) claim that HR departments need to be more strategic when facing virtual HR challenges with the aim of adapting to the uncertainty of technological and market changes. Collaboration, flexibility and change management are underlying requisite competencies detected in this period to cope with technological changes (Lepak and Snell 1998). In this vein, Dyer (1999) identifies a list of competencies for HR managers, such as being a business partner, having technological competencies for the HR function, managing change and competencies in organisational development; his study analyses whether different university programmes were preparing human resource professionals for their future. Particularly, Huston (2008) highlights a set of requisite competencies for nurse leaders, such as having a global mindset in their field, technological skills, decision-making skills, ability to create organisation cultures, comprehension and intervention in political issues and ability to develop collaborative skills, among others. Requisite competencies go beyond mere technologically-driven aspects, emphasising the need to empathise with the workforce. This fact is considered at both the educational and the organisational levels. Indeed, human resource managers have acknowledged the variety of competencies required to deal with the workforce, with people representing the primary source of competitive advantage (Rowley and Warner 2007). In this line, an interesting approach to competencies in the twenty-first century within organisations was proposed in the Special Issue on this topic in the Journal of Management Development, in which the guest editors stated that *emotional, social and cognitive intelligence competencies predict effectiveness in professional, management and leadership roles in many sectors of society* (Boyatzis 2008). In addition, numerous studies published in this period are related to different education programmes addressing business management, health management and technology management, and emphasising value creation-learning (Anderson 2010; Huston 2008; Kerr and Lloyd 2008). Lawler and Elliot (1996) study an AI tool to help with HR management decision

making, finding an expert system that can replicate some non-trivial problem solving competencies in HRM. According to Miles et al. (2000), innovation stems from an underdeveloped skill, namely collaboration. Organisations that understand how to collaborate better create and transfer knowledge, and this knowledge can lead to innovation. Innovation and knowledge are key factors for future organisations.

5.1.2 2011–2015

From 2011 to 2015, *big data* and *human resource management* emerged as motor themes for the competencies for AI field of study (Fig. 7, see “Appendix”). In this period, the *big data* research theme appears as an independent cluster with related research subthemes, such as *education*, *IT*, *innovation* or *knowledge*. According to the Chartered Institute of Personnel and Development report, big data will enable the HR function to leverage and capture the important information. Big data early adopter companies have faced significant challenges, such as difficulties in obtaining the technical skills to support big data tools; differences in the supply of workers with the requisite skills may explain the differences in the adoption of IT innovations (Tambe 2014). Big data and talent analytics have been considered as important capabilities for the HR function (CIPD 2013). Preparing new generations with skills such as creativity and technical capabilities to compete in the twenty-first century is a significant concern for country innovation systems, so education policies must be integrated in the national innovation strategy (Ibata-Arens 2012). In addition, requisite leadership competencies such as decisiveness, proactivity, innovative decision making and intelligent stewardship are aimed at obtaining agile, flexible and cross-cultural responses to lead effectively and ethically in a changing and globalised environment (Sheppard et al. 2013). These authors emphasise the use of leadership development programmes based on action learning or mentoring to develop these skills. Twenty-first century developments and challenges such as big data and network activities have major implications for HRM models beyond the mere single employer–employee relationship (Swart and Kinnie 2014). These authors identify three HRM models (i.e., buffering the network, borrowing from the network, and balancing the network), which are suitable to enhance networked working grounded on new technological developments. Overall, an increasing interest in employees’ skills, leadership skills and best practices from organisations and educational institutions was observed throughout this period (Sheppard et al. 2013; Tambe 2014).

5.1.3 2016–2020

In the last period (2016–2020), *data science* and *firm performance* are motor themes, *artificial intelligence*, *innovation* and *future* are basic themes, *competence*, *outcomes* and *self-efficacy* are specialised themes and *analytics* is an emerging or disappearing theme (both weakly developed and marginal to the field) (Fig. 3). As most of the publications belong to this period (Fig. 5, see “Appendix”), we analyse it in depth in this section. Table 7 (see “Appendix”) shows the performance measure analyses with the number of documents, citations and *h*-index per theme.

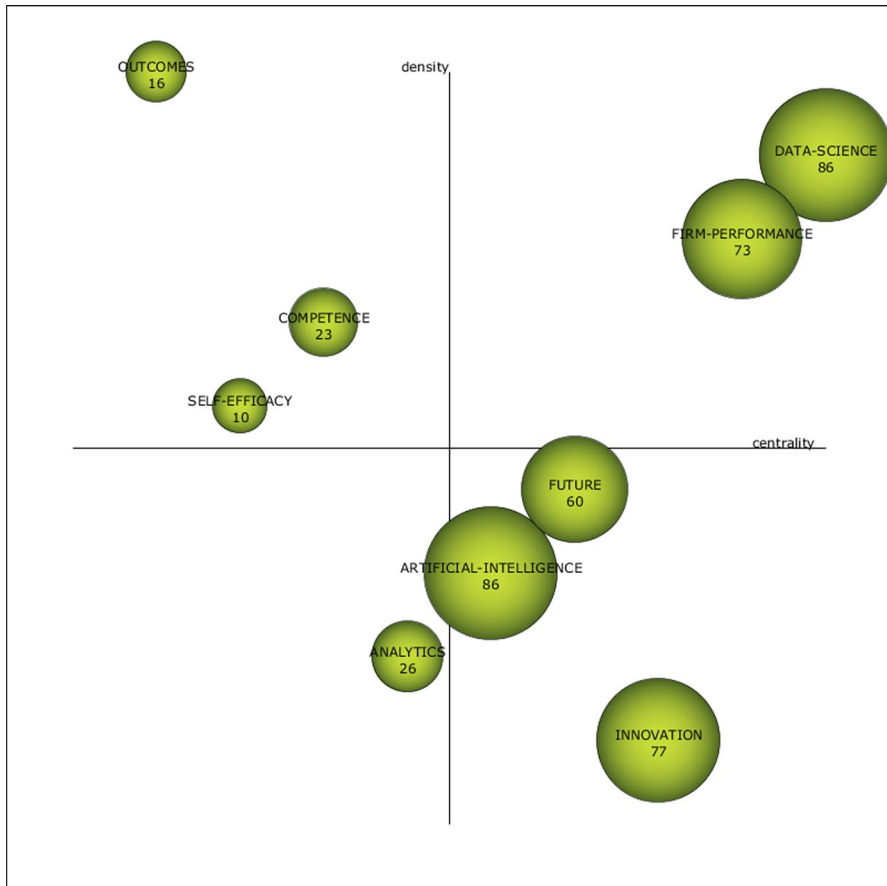


Fig. 3 Strategic diagram of the competencies and skills for AI applications' themes from 2016 to 2020 (number of union documents)

Data science, with 86 documents and 611 citations, covers research subthemes such as *business intelligence*, *predictive analytics*, *data quality* and *knowledge management* (Table 7, see “Appendix”). The enormous variety, volume and generation rate of data available can be properly channelled by the data analytics competency to improve a firm’s decision-making performance (Ghasemaghahi et al. 2018). The data analytics competency is a five-dimension formative index defined as the *firm’s ability to deploy and combine data analytics resources for rigorous and action-oriented analyses of data* (Ghasemaghahi et al. 2018; p.103). Rialti et al. (2019) state that big data may impact a company’s performance by influencing its capability and adaptability, and they refer to the need to develop organisational big data analytics capabilities (i.e., infrastructure flexibility, management capabilities and personnel capabilities) to obtain significant information for decision making. Thus, data science has quickly emerged in both the traditional business models and knowledge

management within organisations, fostering the growth of new competencies such as data analytics.

Firm performance, with 73 documents and 452 citations, is another motor theme in this field of study (Table 7, see “[Appendix](#)”). *Firm performance* covers topics such as the *resource-based view* and *supply chain management*, among others. A number of studies analyse the influence of big data and predictive analytics on company performance, such as supply chain, operational and healthcare performance (Dubey et al. 2019; Peeters et al. 2020; Rialti et al. 2019; Wang et al. 2019). Dubey et al. (2019) study the influence of external pressures on the organisational resources moderated by the impact of big data capability, explaining how this capability influences the operational and cost performance of the organisation. Wang et al. (2019) examine the complexity of big data analytics within healthcare organisations, exploring how big data analytics act with firm resources and capabilities in diverse configurations to improve the quality of care. Peeters et al. (2020) develop a people analytics effectiveness wheel, in which the people analytics team need to focus on four aspects: resources, products, stakeholder management and governance structure. Further research on how big data contributes to firm performance is required. The *analytics* research topic is an emerging theme and covers subthemes such as *information* and *e-HRM*. Indeed, the new developments incorporated to the e-HRM and information subthemes enhance organisational performance. AI can be applied to obtain diverse information from social media for organisations to use (Kaplan and Haenlein 2019) and it is a valuable tool for decision making (i.e., recruitment, compensation, etc.). Furthermore, Bondarouk and Brewster (2016) conceptualise the IT and HRM literature to clarify the pros and cons for different stakeholders of the intersection of these fields of study, HRM and technology, traditionally defined as e-HRM, i.e., the performance of HRM activities through the support of channels grounded in web-technologies (Ruël et al. 2004). Thus, AI offers opportunities to control, manage and govern job processes and tasks efficiently and will change the way in which HRM specialists work (Bondarouk and Brewster 2016).

The *future* research theme, together with the *artificial intelligence* and *innovation* transversal clusters, present a large number of publications (Table 7, see “[Appendix](#)”). The *future* cluster is a basic or transversal theme with 60 documents and 307 citations and concerns themes such as *the fourth industrial revolution*, *blockchain*, *Industry 4.0*, *service robots*, *big data*, *smart factory* and *automation*. Another basic or transversal topic is the *innovation* theme, which has 77 documents and 437 citations and covers subthemes such as *learning*, *strategic HRM* and *intelligence*. The *artificial intelligence* cluster, with 86 documents and 514 citations, is a basic or transversal theme and discusses subthemes such as *machine learning*, *robotics*, *digitalisation* and *human resource management*. Indeed, the arrival of Industry 4.0, or the fourth industrial revolution, focuses on technology such as artificial intelligence and advanced robotics, going beyond a mere technological challenge to also include a human challenge (Rampersad 2020; Santana and Cobo 2020). To survive in the future, human skills and competencies to effectively work and cope with new technological developments are crucial (Rampersad 2020). Kaplan and Haenlein (2019) present a Three C Model to support firms with the internal and external challenges of artificial intelligence, which they define as Confidence, Change and Control.

Furthermore, Ghobakhloo (2018; p.910) provides a strategic roadmap for manufacturers transitioning toward Industry 4.0, which is *an integrative system of value creation comprised of 12 design principles and 14 technology trends*. According to this author the Industry 4.0 roadmap is not a one-size-fits-all suitable for every company strategy and should consider the firms' core competencies, motivations, targets, capabilities and budgets. The human-machine interaction is an important concept in this *artificial intelligence* research topic. Klumpp (2018) develops a multidimensional conceptual model to differentiate the performance of human-artificial collaboration systems before investment decisions. This author identifies four levels of resistance before an efficient and trusted collaboration is attained between humans and artificial intelligence systems. To overcome many of the challenges that artificial intelligence poses to human resource management, Garcia-Arroyo and Osca (2021) highlight the management of resistance to change within organisations through data experts or multidisciplinary teams to support HRM, and the convenience of alliances between firms and educational institutions. These aspects are also related to the specialised research theme, *self-efficacy*, covering subthemes such as *acceptance*, which refers the acceptance of these new technologies by the organisations, but also by the end-users (Kaše et al. 2019; Martínez-Caro et al. 2018).

The study of the requisite competencies and skills to cope with these artificial intelligence challenges within organisations constitutes a specialised theme, together with the *outcomes* and previously analysed *self-efficacy* research themes. The *competence* cluster covers research subthemes such as *abilities* and *success*. The *outcomes* cluster discusses *satisfaction* and *work-engagement* subthemes in the AI age. Regarding the *compentece* cluster, it is known that employees need to be innovative, detect opportunities and possess certain competencies to overcome the fear that robots and AI will replace many positions (Rampersad 2020) in a wide range of industries such as tourism and culture. Despite the increasing concern about the gap between existing and requisite employees' digital competencies, more effort is needed to face the challenges of digitalisation in the twenty-first century (Oberländer et al. 2020). Sousa and Rocha (2019) conduct interviews to find out which skills managers require to cope with new disruptive technologies and list the following skills according to three dimensions: innovation, leadership and management. In turn, Sousa and Wilks (2018) detect critical competencies (i.e., critical thinking and problem solving, collaboration in networks and leading by influence, agility and adaptability, initiative and entrepreneurship, effective oral and written communication, evaluating and analysing information, and curiosity and imagination) and technological disruptive competencies (i.e., artificial intelligence, nanotechnologies, robotisation, the internet of things and augmented reality).

Van Laar et al. (2017) make a significant contribution by highlighting the digital skills needed for the twenty-first century, namely: technical competencies, information management, communication, collaboration, creativity, critical thinking and problem solving. Murawski and Bick's (2017) research considers the following selected digital competencies: information processing, communication, content creation, safety, problem solving, digital rights, digital emotional intelligence, digital teamwork, making use of big data, self-disruption, making use of artificial intelligence and virtual leadership. Furthermore, these authors call on organisations to

focus on *the alignment of multiple stakeholders for the design of ‘digital’ curricula and the integration by HR departments of the construct of digital competences, e.g. for compensation matters and job requirements* (Murawski and Bick 2017; p.721).

HR professionals must decide what types of capabilities are needed and whether to create their own capabilities or buy them in. Despite the possibilities of big data for the HR function, Angrave et al. (2016, p.1) are critical of the current approach to HR analytics, stating that the HR function must engage *operationally and strategically to develop better methods, since it is unlikely that existing practices of HR analytics will deliver transformational change*. In turn, Alharthi et al. (2017) also consider that organisations are not ready to make use of the big data capabilities, listing a number of recommendations to address big data barriers. For example, technological barriers can be infrastructure readiness and complexity of data, human barriers can be privacy and a lack of skills, and an organisational barrier can be the organisational culture (Alharthi et al. 2017; Manyika et al. 2011). With respect to human barriers, Alharthi et al. (2017) indicate that organisations need to collaborate with educational institutions in order to develop or acquire the requisite skills for the big data era, and that organisations should include privacy protection measures to enhance existing processes related to big data. In the 2014–2017 period, it became clear that it is crucial to leverage AI through HR.

In addition, Sousa and Wilks (2018) state that critical and technological disruptive competencies are needed to obtain sustainable employability. In this vein, Perriñez-Cañadillas et al. (2019) find that candidates’ digital competencies of communication, content creation, safety and problem solving are determinant in the selection decision. In conclusion, competency mismatch is a fact (Acemoglu and Autor 2011; Cukier 2019; Oberländer et al. 2020), but work-integrated learning and technology-enabled talent matching platforms can reduce the effect of new technological developments, and efforts need to be made through specific training from educational institutions and organisations (Alharthi et al. 2017), as previously stated. In this line, Di Gregorio et al. (2019) propose an integrated model of employability competencies in the marketing field (e.g., soft competencies, analytical competencies, digital and technical competencies, core marketing competencies and customer insight competencies) to be implemented in universities that will help marketing graduates succeed in the digital domain, since related marketing professions such as e-commerce manager, social media manager, digital marketing manager or big data analyst have been signalled as key future positions.

5.2 Evolution map

In this section, we examine the evolution of the different research themes in the sub-periods 1992–2010, 2011–2015 and 2016–mid-2020 (Fig. 4). The academic themes identified in each period were observed with SciMAT based on the evolution of the keywords over the years. From 1992 to 2010, the incipient field of study into competencies and skills for AI applications mainly covered research themes related to the acquisition and learning of the requisite resources and competencies to increase knowledge within organisations and educational institutions (Dyer 1999; Lawler and

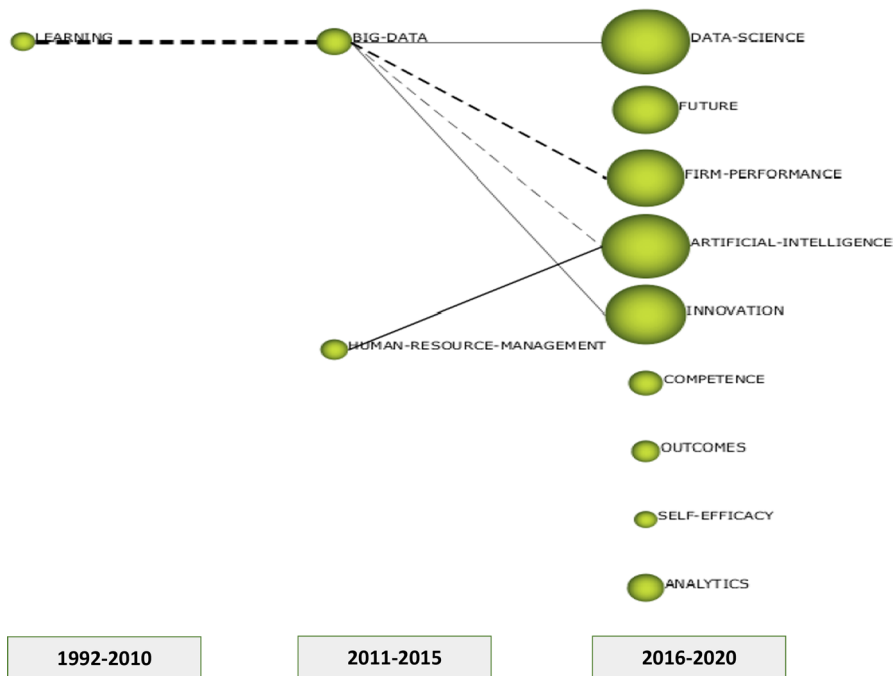


Fig. 4 Evolution map of the competencies and skills for AI application themes for the periods 1992–2010, 2011–2015 and 2016–2020

Elliot 1996; Lepak and Snell 1998). In the period 2011–2015, the *learning* theme evolved into the *big data* research topic, as real proof of the growing importance of the big data applications in business and coinciding with the incorporation of the term big data in the Oxford English Dictionary in 2013 (Press 2013; Wasserman 2013). The *human resource management* cluster emerges for the first time as an independent research theme, highlighting the need to leverage artificial intelligence through human resource management. In the last period (2016 to mid-2020), new clusters appear such as *data science*, *artificial intelligence*, *future* and *firm performance* are derived from the *big data* research theme, reflecting the importance of the development of AI within organisations and the need for efficient management. *Competence* is seen for the first time as a separate topic, with a steadily increasing number of authors studying the requisite competencies needed to face the artificial intelligence era, such as analytics and information interaction (Marler and Boudreau 2017; Sun et al. 2017). Finally, the research themes *self-efficacy* (related to the *user acceptance* subtheme) and *outcomes* (associated with work-engagement and satisfaction) appear as separate themes in this period, discussing the challenges that AI users currently encounter. This conceptual evolution map shows human resource management as an important resource for enabling the requisite competencies, together with the educational institutions, to face the development of AI within organisations.

6 Discussion and conclusions

6.1 Theoretical contribution

The rapid growth of AI has brought associated changes in both organisations and society through its application in both personal and professional spheres. At the organisational level, AI influences employees and human resource management and presents two major challenges: how to ensure that human resources are able to use the full potential of AI in their jobs and how AI influences human resource management.

This research has detected certain HRM practices that enhance the full potential of AI in jobs. Selecting, planning and training choices are amongst the HRM practices most frequently identified to leverage AI. Organisations that take into account the challenges of AI when investing in skills development and planning ahead are predicted to respond better (Gallardo-Gallardo and Collings 2021). According to these authors, organisations should rely on competencies rather than job descriptions or traditional roles; thus competencies are the foundation on which to develop careers, redesign jobs, deploy talent and evaluate employees. Our study has identified the United States and the United Kingdom as the most prolific countries in this field of study, which mirrors the fact that the most technologically advanced companies (e.g. Apple or Alphabet) are predominantly from these countries, and have traditionally focused on underlying competency sets in assessing their employees' readiness (Cappelli and Keller 2017). In addition, training in organisations and educational institutions is needed to reduce competency mismatch (Acemoglu and Autor 2011; Alharthi et al. 2017); talent mobility in particular is a source of knowledge for employees that bolsters competencies (Khilji et al. 2015). Thus, outdated human resource management practices need to be replaced with customised responses to attract and develop digital talent in a manner that encourages employees with these skills to stay in the organisation (Cantoni et al. 2018; Laker 2022; Malik et al. 2020a, b). Laker (2022) makes some recommendations to retain and develop digital talent: (1) know employees well (listen to them carefully and learn what motivates them); (2) act with honesty from hiring to motivation and compensation; (3) customise HRM practices in accordance with talented employees' competencies; and (4) adopt particular responses to support well-being and work-life balance. These HR measures will help employees to harness the full potential of AI in their jobs.

This review of the different periods and the evolution of research in the AI and digital competencies field provides an extensive overview of the digital competencies field. As previously noted, most of the literature on digital competencies has focused on employees' competencies in general, although recent studies also include the specific competencies that HR practitioners need to manage new HR applications incorporated into HRM, and the leadership competencies to lead in this digital environment. Indeed, Malik et al. (2020a, b) state that both analytical and tech-savvy

competencies are key competencies for HR practitioners; they need to be more analytical when looking at available HR data to inform their decisions, and they need to be familiar with available technologies, such as having a working knowledge of certain programming languages. These analytical and tech-savvy competencies will enhance HR practitioners' performance. Similarly, Cortellazzo et al. (2019) identify the main competencies leaders need in the digital transformation era: communicating through digital media, high-speed decision making, managing disruptive change, managing connectivity, and the renaissance of technical competencies. Boyatzis (2008, p.5) states that *emotional, social and cognitive intelligence competencies predict effectiveness in leadership roles*. Leaders therefore need to understand how their organisations are influenced by the ubiquitous digital technology. They need to ask the right questions, respond to exceptional situations informed by algorithms and act beyond machines (Dewhurst and Willmott 2014).

When social researchers refer to the influence of technology on HRM, the term "digital" is usually applied to refer to all types of technological developments such as AI, machine learning, internet of things or big data, but also to internet. However, the crucial digital revolution has come from AI and its implementation within organisations and society in general. Data analytics or big data competencies are essential to face AI challenges (Ghasemaghahi et al. 2018; Rialti et al. 2019), but human competencies (e.g., critical thinking, problem solving, communication and teamwork) to effectively work and cope with new technological developments are vital for the development of innovation essential in the AI era (Rampersad 2020). Indeed, as Ferrari (2012) states, digital competencies cover much more than technical skills. The different classifications of digital competencies reviewed in this systematic study are shown in Table 2.

In consequence, certain digital competencies must be brought together in this Industry 4.0, such as problem solving, non-routine tasks and creation of digital outputs (Djumaieva and Sleeman 2018), since some technologies, such as the Internet of Things (IoT), big data and AI, will automate many of the HR processes, resulting in small and efficient HR teams. Some authors also consider the huge rapid transformation taking place in all areas. Murawski and Bick (2017) propose a typology of digital competencies that brings together the digital competencies listed by the European Commission in the DIGCOMP project and other competencies that have been identified as necessary in today's digital context such as virtual leadership, digital emotional intelligence or making use of big data, among others.

Another classification separates competences into cognitive and meta-cognitive competencies (critical thinking, creative thinking, learning to learn and self-regulation), non-cognitive competencies (empathy and collaboration) and digital competencies (use of digital devices) (OECD 2018). Non-cognitive competences include: open-mindedness, openness to learn and change, flexibility, curiosity, innovation, creativity, entrepreneurship, resilience, planning/organisation, responsibility, persistence, teamwork, communication, initiative, sociability, empathy, collaboration, emotional control and positivity (Gonzalez-Vazquez et al. 2019; Kautz et al. 2014).

Table 2 Summary of digital competencies

Author	Digital competencies
Ferrari (2012)	Information management, collaboration, communication and sharing, creation of content and knowledge, ethics and responsibility, evaluation and problem solving, technical operations
Murawski and Bick (2017)	Information processing, communication, content creation, security, problem solving, digital rights, digital emotional intelligence, digital teamwork, making use of Big Data, self-interruption, making use of artificial intelligence and virtual leadership
van Laar et al. (2017)	Technical competencies, information management, communication, collaboration, creativity, critical thinking and problem solving
OECD (2018)	Cognitive and meta-cognitive skills (e.g. critical thinking, creative thinking, learning to learn and self-regulation); social and emotional skills (e.g. empathy, self-efficacy and collaboration); and practical and physical skills (e.g. using new information and communication technology devices)
Sousa and Wilks (2018)	Critical skills (i.e., critical thinking and problem solving, collaboration in networks and leading by influence, agility and adaptability, initiative and entrepreneurship, effective oral and written communication, evaluating and analysing information, and curiosity and imagination) and technological disruptive skills (i.e., artificial intelligence, nano-technologies, robotisation, internet of things and augmented reality)
Gonzalez-Vazquez et al. (2019)	Information and data literacy, communication and collaboration, media literacy, digital content creation (including programming), safety (including digital well-being and competences related to cybersecurity), intellectual property related questions, problem solving and critical thinking
Oberländer et al. (2020)	Handling of hardware, handling of software, programming, handling of applications, innovative capability and creativity, information processing (recognising one's own knowledge gaps, search, data analysis, evaluation), data organisation, effective usage, communication, collaboration, networking, netiquette, sharing data with others, cultural aspects, security law, responsibility, goals and motivation, willingness to learn and openness, ethics and morals, autonomy and Independence, problem solving, training/educating others

As with any type of competency, the classifications of digital competencies are multiple and there is still no consensus on them. However, a review of the literature on AI and competencies allows us to identify those that are most needed to leverage AI in organisations. Specifically, these are communication and problem solving, followed by collaboration, teamwork and technical skills, which include all those aspects related to programming and the use of software and hardware tools linked to AI and big data.

6.2 Managerial implications

Organisations need to acknowledge the diversity of digital competencies for AI, which can be technical, methodological, social and personal (Hecklau et al. 2016). The social and technological aspects do not innovate in isolation. This perspective assumes that it is necessary to develop new approaches to work in an integrated way with qualifications and organisational learning instruments, especially on digitisation (Schuh et al. 2015). Moreover, collaborative types of learning and learning environments are critical for organisations to operationalise knowledge and skills, such as virtual learning environments (VLEs) (Richert et al. 2016), augmented reality (Azuma 1997) and collaborative environments between cobots (service robots) and humans (Calitz et al. 2017).

Furthermore, organisations should be aware of the difficulties they must overcome to ensure their workforces really do have the competencies required to ensure AI contributes to their competitive advantage. Some of these obstacles are outdated IT infrastructures, the inherent complexity of AI itself, the lack of the necessary competencies in employees and organisational cultures that do not foster the development and implementation of AI (Alharthi et al. 2017). Overcoming these difficulties requires employees to acquire and develop the necessary competencies, as well as updating the IT infrastructure while implementing new forms of management and a new organisational culture (Manyika et al. 2011).

The increased importance of AI and big data has led organisations to increase their demand for data scientists, data analysts, and experts in AI and big data technology, among others. Organisations should collaborate with educational institutions to align curricula with industry requirements in terms of competencies related to AI and big data, that is, digital competencies (Alharthi et al. 2017). Some universities are already working in this line to meet the industry needs and are training students in technologies related to AI and big data as well as analytical skills (Di Gregorio et al. 2019; Miller 2014). First of all, it is necessary to know how AI and big data technologies work. In addition, technical competencies are required, such as knowledge and advanced skills in mathematics and statistics, machine learning, predictive analytics, decision-making models and data visualisation, which must be incorporated into training programmes (Miller 2014). Thus, it is important for organisations to collaborate with educational institutions to develop big data competencies and include privacy protection actions to improve big data processes (Alharthi et al. 2017).

Moreover, it is also important for employees to acquire general analytical competencies, such as research, communication and problem solving, among others. The reason for this is that, in a future characterised by increased automation, advanced robotics, AI, rapid change and industry transformation, the next generation of employees must have not only the competency of problem solving, but also critical thinking, communication, teamwork, innovation and entrepreneurship (Täks et al. 2014). Previous studies point to the increasing demand for a set

Table 3 Research agenda for the AI-digital competencies–HRM literature

Research themes	Author	Research agenda
Data science, firm performance	Ghasemaghaei et al. (2018)	To validate and operationalise the bigness of data construct (of the Data Analytics Competency index) considering other suggested big data characteristics (such as value or veracity)
Artificial intelligence, self-efficacy	Klumpff (2018)	To study how to increase/reduce levels of acceptance/resistance through information and training or experience on the human collaborators' side
Competence	Sousa and Wilks (2018)	To detect and analyse the processes that can be used to enhance the skills required to meet the demands of the digital age
Data science, firm performance	Rialti et al. (2019)	To search for the antecedents of organisational big data analytics BDA capabilities, studying other elements such as ambidexterity, agility and performance
Data science, firm performance	Dubey et al. (2019)	To analyse the impact of BDA capabilities on firms' performance using qualitative methods
Artificial intelligence and HR	García-Arroyo and Osca (2021)	To conduct more research on the Big Data and Predictive Analytics and Manufacturing Performance relationship over a longer period with samples from more industries, countries and informants with more diverse backgrounds
Firm performance	Peeters et al. (2020)	To observe examples of good practices in the handling of technological or methodological challenges when implementing big data systems in HRM (especially selection and hiring, and in assessment and development), and the application of new studies that analyse the multilevel and the time-level effect of big data in HR practices
Artificial intelligence	Rampersad (2020)	To use the People Analytics Effectiveness Wheel as a guideline in future qualitative studies, paying specific attention to stakeholders and governance
Competence	Oberländer et al. (2020)	To study the relationship between the key ingredients (included in the enablers, governance, stakeholder management and deliverables categories) needed to set up an effective people analytics team
		To further study the impact of AI and big data on innovation
		To use the validated quantitative tool to obtain feedback about the work-integrated learning programme and control effectiveness of education programmes
		To further study how the different dimensions of the digital competencies are interrelated
		To analyse in depth the dynamic frameworks of digital competencies at work that consider future-oriented job requirements

of higher education competencies, such as soft competencies and creative intelligence (Velthuijsen et al. 2017).

6.3 Research agenda

Our extensive and objective bibliometric analysis has revealed some future research lines in several research themes (Table 3). For the purpose of this paper, these research lines are divided into three main groups.

Research line 1: AI-digital competencies and firm performance The study of variables related to digital competencies and firm performance is increasingly important (Dubey et al. 2019; Peeters et al. 2020; Rialti et al. 2019). These authors highlight the need to study issues related to *AI-digital competencies and firm performance* such as: (1) the big data and predictive analytics and manufacturing performance relationship over a longer period with samples from more sectors, countries and informants with different backgrounds; (2) the use of the People Analytics Effectiveness Wheel as a guideline in future qualitative studies, particularly in observing the influences of stakeholders and governance; and (3) the impact of big data capabilities on firms' performance using qualitative methods.

Research line 2: the impact of AI and big data on diverse variables such as HR or innovation Some research focuses on the *AI* research themes (Garcia-Arroyo and Osca 2021; Rampersad 2020). In this line, Rampersad (2020) calls for more studies on the impact of AI and big data on innovation, and Garcia-Arroyo and Osca (2021) would like to see more examples of good practices in the management of technological or methodological challenges when implementing big data systems in HRM (especially selection and hiring, and assessment and development), and new studies that analyse the multilevel and the time-level effect of big data in HR practices. Given the lack of studies on AI application in the selection, development and assessment processes (Garcia-Arroyo and Osca 2021), more research is needed on the ethical implications for the workforce of this advancement. Important questions to address the AI challenges facing HRM include understanding how HR departments should manage employees' private data or how to protect data through current regulations that guarantee the rights of employees.

Research line 3: self-efficacy, level of acceptance of/resistance to AI developments The *self-efficacy* research theme (including the *acceptance cluster*) in AI developments by users is also attracting scholars' attention. Klumpp (2018) calls for further investigation on how to increase/reduce levels of acceptance/resistance through information and training or experience from the human collaborator perspective. Finally, in relation to the academic theme of *skills* or competencies, Sousa and Wilks (2018) identify a need to detect and observe the processes that can be used to enhance the requisite skills to meet the demands of the digital age, and Oberländer et al. (2020) recommend further study on how the different dimensions of the digital competencies are interrelated.

6.4 Limitations

The main limitation of the study is the restriction of our analysis to WoS articles and reviews from the business and management areas, although we consider that they offer an optimal representation of the AI-competencies–HRM research field. A second limitation is related to the fact that the quality of the results depends on the selected keywords and the scope of the database, as many journals' reference data do not contain keywords, particularly in the earliest years, and some documents may not include all the important aspects of the text in their keywords (the so-called indexer effect). However, to improve the quality of the data, this study also used title and abstract information in addition to the keywords. A third limitation is that this research area is an incipient but rapidly growing field of study (as demonstrated by the increasing number of publications on the topic). Nevertheless, this limitation is also a future potential strength: as this field of study develops, we expect to see an increase in research on AI-digital competencies–HRM in the coming years.

7 Conclusions

This bibliometric study explored employees' digital competencies to face AI developments with an approach from the HRM side. We observed that, in the last analysed period (2016–2020), the topics were classified as motor (*data science, firm performance*), basic (*artificial intelligence, future and innovation*), specialised (*self-efficacy, competence and outcomes*) and emerging (*analytics*).

In summary, we conclude that the impact of AI on organisational performance will only be effective if employees have the appropriate digital competencies to achieve the fit between AI and organisational performance. We hope that this study will serve as a systematisation and representation of the relationship between employee competencies and AI and also as the basis for further studies, especially on motor and basic themes. For instance, what type of training can enhance digital competencies? How can employees' digital competencies improve firm performance? Some studies have recently started to delve into these topics, but more research is needed.

AI has clearly entered our lives and will have an impact on society and companies. Therefore, an in-depth analysis of the literature on the competencies for implementing and leveraging artificial intelligence and its impact on HRM is needed. It is essential to adapt the current and future workforce to ensure that employees will be innovative and capable of detecting opportunities that transform industries and provide creative solutions to achieve global objectives. It is therefore crucial for employees to acquire and develop the competencies demanded by firms, especially digital competencies.

Appendix

See Tables 4, 5, 6, 7 and Figs. 5, 6, 7.

Table 4 Highest cited publications and authors in the competencies and skills for the AI scientific domain

	Title	Author	Year	Citations
1	Data science, predictive analytics, and big data: a revolution that will transform supply chain design and management	Fawcett, SE, Waller, MA	2013	369
2	Case study on tour guiding: professionalism, issues and problems	Ap, J, Wong, KKF	2001	165
3	Virtual HR: strategic human resource management in the twenty-first century	Lepak, DP, Snell, SA	1998	153
4	Technological learning, knowledge management, firm growth and performance: an introductory essay	Hitt, MA, Ireland, RD, Lee, HU	2000	152
5	Data science, predictive analytics, and big data in supply chain management: current state and future potential	Schoenherr, T, Speier-Pero, C	2015	101
6	The Future.org	Miles, RE, Snow, CC, Miles, G	2000	97
7	The making of twenty-first-century HR: An analysis of the convergence of HRM, HRD, and OD	Ruona, WEA, Gibson, SK	2004	94
8	Artificial intelligence in service	Huang, MH, Rust, RT	2018	92
9	The organisation of the future: strategic imperatives and core competencies for the twenty-first century	Nadler, DA, Tushman, ML	1999	87
10	Big data investment, skills, and firm value	Tambe, P	2014	76
11	Preparing nurse leaders for 2020	Huston, C	2008	72
12	The future of manufacturing industry: a strategic roadmap toward Industry 4.0	Ghobakhloo, M	2018	66
13	Are we there yet? What's next for HR?	Ulrich, D, Dulebohn, JH	2015	66
14	Shifting intra- and inter-organisational innovation processes towards digital business: An empirical analysis of SMEs	Santoro, G, Scuotto, V, Bresciani, S, Del Giudice, M	2017	63
15	Emerging patterns of complex technological innovation	Kash, DE, Rycroft, R	2002	61
16	eLancing: a review and research agenda for bridging the science-practice gap	Aguinis, H, Lawal, SO	2013	57
17	Siri, in my hand: Who's the fairest in the land? On the interpretations, illustrations, and implications of artificial intelligence	Kaplan, A, Haenlein, M	2019	51
18	HR and analytics: why HR is set to fail the big data challenge	Angrave, D, Charlwood, A, Kirkpatrick, I, Lawrence, M, Stuart, M	2016	50

Table 5 Most prolific journals in the competencies and skills for the AI scientific domain

	Journal	Number of documents	5-Year impact factor
1	INTERNATIONAL JOURNAL OF HUMAN RESOURCE MANAGEMENT	12	3.8
2	JOURNAL OF MANAGEMENT DEVELOPMENT	10	ESCI
3	QUALITY-ACCESS TO SUCCESS	8	ESCI
4	HUMAN RESOURCE MANAGEMENT REVIEW	8	5.450
5	MANAGEMENT DECISION	8	2.886
6	HUMAN RESOURCE MANAGEMENT	8	3.702
7	TECHNOLOGICAL FORECASTING AND SOCIAL CHANGE	7	5.179
8	BUSINESS HORIZONS	7	4.311
9	BUSINESS PROCESS MANAGEMENT JOURNAL	6	2.121*
10	INTERNATIONAL JOURNAL OF LOGISTICS MANAGEMENT	6	3.289
11	INTERNATIONAL JOURNAL OF MANAGEMENT EDUCATION	5	2.354*
12	INTERNATIONAL JOURNAL OF EDUCATIONAL MANAGEMENT	5	ESCI
13	JOURNAL OF ORGANIZATIONAL EFFECTIVENESS-PEOPLE AND PERFORMANCE	5	ESCI
14	JOURNAL OF BUSINESS LOGISTICS	4	5.877
15	INTERNATIONAL JOURNAL OF MARKET RESEARCH	4	1.466
16	JOURNAL OF GLOBAL RESPONSIBILITY	4	ESCI
17	BRITISH JOURNAL OF MANAGEMENT	4	4.372
18	JOURNAL OF STRATEGIC INFORMATION SYSTEMS	4	6.697
19	TECHNOLOGY ANALYSIS & STRATEGIC MANAGEMENT	4	2.105
20	INTERNATIONAL JOURNAL OF TECHNOLOGY MANAGEMENT	4	1.592

ESCI Emerging Source Citation Index

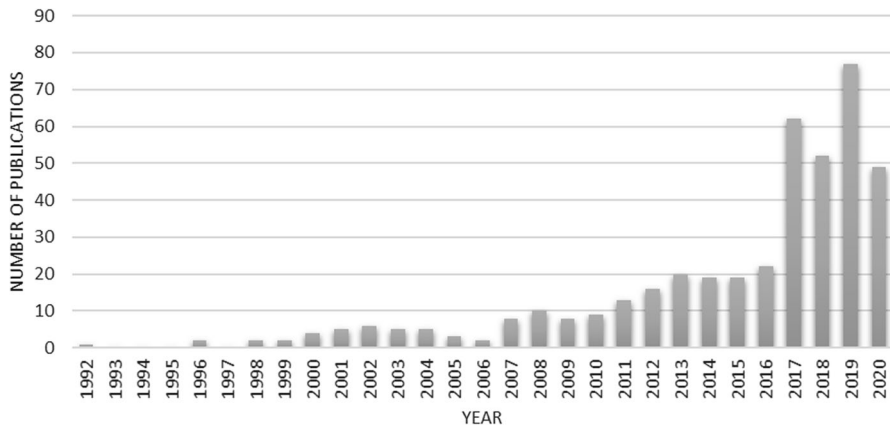
*JCR 2019

Table 6 Most prolific countries in the competencies and skills for the AI scientific domain

	1992–2020	% of total
USA	149	27
UK	66	12
Australia	25	5
Canada	25	5
India	20	4
The People's Republic of China	20	4
Germany	18	3
France	13	2
Spain	12	2
Italy	11	2
South Africa	11	2
Netherlands	10	2

Table 7 Performance of the competencies and skills for AI application themes for the period 2016–2020 (union documents)

Name	Number of documents	Number of citations	<i>h</i> -index
DATA SCIENCE	86	611	13
FIRM PERFORMANCE	73	452	11
FUTURE	60	307	7
ARTIFICIAL INTELLIGENCE	86	514	11
INNOVATION	77	437	11
COMPETENCE	23	215	13
OUTCOMES	16	101	5
SELF-EFFICACY	10	20	3
ANALYTICS	26	152	7

**Fig. 5** Number of WoS publications on competencies for the AI research field from 1992 to mid-2020

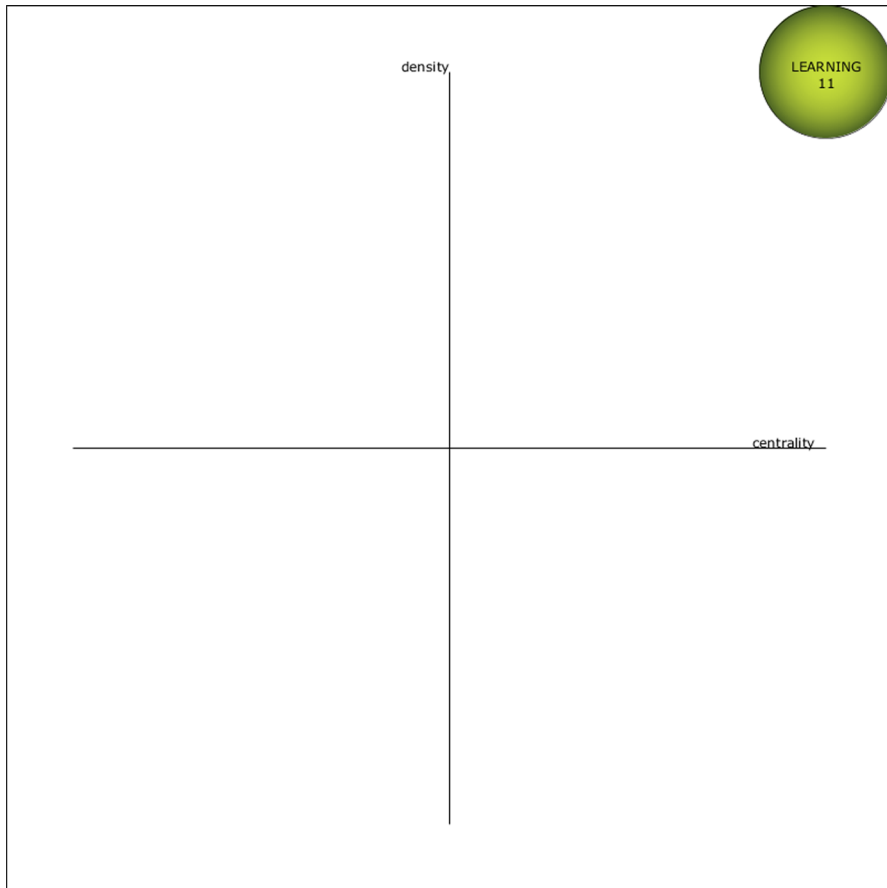


Fig. 6 Strategic diagram of the competencies and skills for AI application themes from 1992 to 2002 (number of union documents)

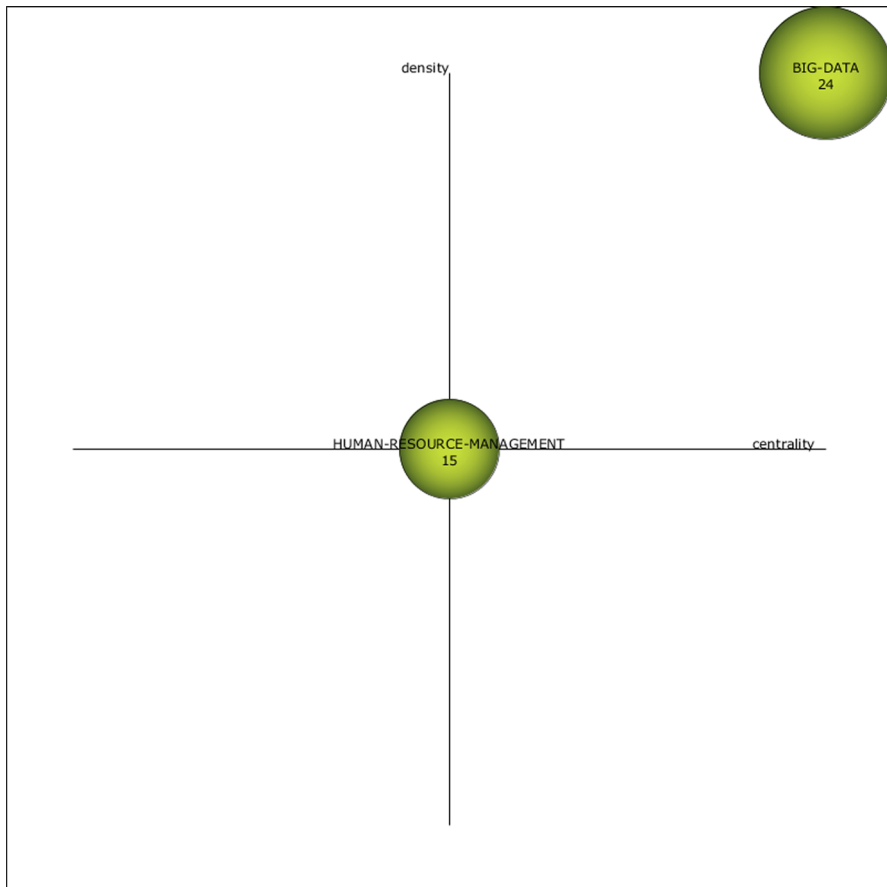


Fig. 7 Strategic diagram of the competencies and skills for AI application themes from 2011 to 2015 (number of union documents)

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